

Penuel Stanley-Zebulon

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EDUCATION

Pennsylvania State University - Harrisburg, PA
B.S. in Artificial Intelligence Methods and Applications (AIMA)

Expected: May 2028

Pan-Atlantic University - Lagos, Nigeria
Bachelor's coursework in Electrical/Electronics Engineering

Nov. 2022 - Jul. 2023

Relevant Coursework: Artificial Intelligence, Statistics, Math for Machine Learning, Object-Oriented Programming, Data Structures and Algorithms, Discrete Math for Computer Science

SELECTED AI SYSTEMS & PRODUCT PROJECTS

Alethia Live - Real-Time Healthcare Navigation Assistant

Mar. 2026

[Live Demo](#) | [GitHub](#) | Technologies: Next.js, TypeScript, Gemini Live API, Google Cloud Run

- Enabled real-time healthcare navigation for patients reviewing after-visit summaries and discharge instructions by deploying a multimodal voice and screenshot assistant with Next.js, TypeScript, Gemini Live API, and Google Cloud Run.
- Mitigated unsafe-answer risk by implementing structured outputs, red-flag awareness, and guardrails that block diagnosis and treatment advice.

Sema - Patient Evidence Packet Agent

Jun. 2026

Technologies: Next.js, React, TypeScript, Tailwind CSS, browser-local voice capture, automated testing, browser PDF export

- Validated Sema's page-aware AI, browser-local voice capture, and deterministic clinician-ready packet generation with 217/217 passing automated checks spanning AI safety, live-agent behavior, voice workflows, and packet boundaries.
- Built an 8-stage review and provenance pipeline with 30 packet-specific tests that blocked stale or unapproved AI content, unsupported medical conclusions, and raw-media leakage; designed to target a 50-70% reduction in packet-review preparation time pending pilot validation.

Neonatal Brain MRI Developmental Age-Bin Classifier

Dec. 2025 - Jan. 2026

[GitHub](#) | Technologies: Python, EfficientNet-B0, transfer learning, Grad-CAM, scikit-learn

- Achieved 87.8% macro-F1 and 87.7% accuracy across 29,516 2D neonatal/infant MRI slices and 14 age classes by fine-tuning EfficientNet-B0 with transfer learning.
- Reduced leakage risk through perceptual-hash grouping and group-based splits, then evaluated per-class F1, confusion matrices, Grad-CAM visualizations, and reproducible artifacts.

WORK EXPERIENCE

IBM AI Experiential Lab - Healthcare Track

Feb. 2026 - May 2026

Healthcare AI Product Builder

- Reduced simulated referral-preparation time by about 40% by building Mamathemba, a full-stack maternal-health referral-readiness prototype with structured case intake, missing-field checks, facility comparison, destination rationale, and editable handoff notes.
- Targeted about 30% fewer missing critical fields and 85%+ handoff completeness in simulated workflows using FastAPI/Python concepts, grounded prompts, healthcare guardrails, and human-in-the-loop review.

AI4ALL

Mar. 2026

Ignite Fellow - Applied AI in Healthcare

- Achieved approximately 90% test accuracy and 88% seizure recall by training and evaluating a 1D CNN on 9,500+ CHB-MIT multichannel EEG windows for seizure vs. non-seizure classification.
- Shipped a reproducible Google Colab and Streamlit demo with GitHub documentation and presented model performance, limitations, and healthcare AI risks at a national AI4ALL virtual symposium.

Penn State Harrisburg Center for Teaching Excellence

Oct. 2024 - Present

Student Media Production Assistant

- Produced accessible instructional media across 100+ faculty/course videos through captioning, audio descriptions, video editing, and course-media preparation with multimedia staff.

SKILLS

- Programming and Data:** Python, NumPy, pandas, SQL basics, Google Colab, Jupyter
- AI/ML:** CNNs, transfer learning, model evaluation, leakage-aware splits, Grad-CAM, scikit-learn, TensorFlow/Keras, PyTorch fundamentals
- GenAI and Agents:** Gemini Live API, LLM workflows, prompt engineering, RAG fundamentals, AI agents, agent/tool workflows, structured outputs, AI safety evaluation, automated test harnesses, guardrails, human-in-the-loop design
- Web and Cloud:** Next.js, React, TypeScript, Tailwind CSS, FastAPI, REST APIs, Streamlit, Git/GitHub, Google Cloud Run

LEADERSHIP & INVOLVEMENT

Student Ambassador - Adobe (Penn State University)

Jan. 2024 - May 2025

- Planned and delivered Adobe Express workshops for Penn State students, teaching creative workflows, project development, and communication skills while improving materials through feedback-driven iteration.

Bible Study Leader - InterVarsity Christian Fellowship/USA

Mar. 2024 - Present

- Lead weekly Bible study discussions, coordinate communication and outreach, and support an inclusive student community focused on belonging, reflection, and faith-based growth.